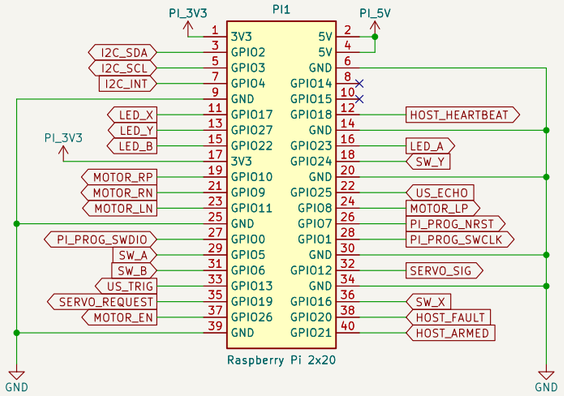
Host GPIO socket (PI1)



Notes

1. Board power enters on header pins 2 and 4 (PI_5V).
2. PI_3V3 feeds only the host-side buffers U6–U9 and U28; the safety domain runs from its own SAFE_3V3.
3. GPIO18 heartbeat (HOST_HEARTBEAT) and GPIO19 servo request (SERVO_REQUEST) belong to the safety interface.
4. GPIO26 (MOTOR_EN) only requests actuator power; the hardware gates on sheet 4 (Independent safety supervisor) decide.
5. GPIO20 (HOST_FAULT) HIGH = healthy; GPIO21 (HOST_ARMED) HIGH = motors armed.
6. GPIO0, GPIO1 and GPIO7 carry SWD and reset while SW5 is at PROG; the host keeps SPI0 and the ID EEPROM probe disabled.
7. GPIO14 and GPIO15 (UART) are not connected.
8. The host mounts inverted on PI1 (pin 1 inward).

Test access

Bare 1.5 mm pads on the bottom side, on the 2.54 mm grid of PI1; one pogo-pin plate reaches all 28 pads.

Signal and programming probe access

Sheet 10

Probe pads in six rows on the bottom side: 24 here; row GND pads TP24–TP26 and TP47 on sheet 11 (Local probe ground returns)
Expected readings for production test
SWD pads TP39–TP43 for the safety MCU

File: testpoints.kicad_sch

Local probe ground returns

Sheet 11

Row GND pads TP24, TP25, TP26 and TP47 (I2C, MOTOR, STATUS and SERVO rows)
TP6, TP40 (POWER, SWD rows): sheet 10 (Signal and programming probe access)

File: ground_probes.kicad_sch

PI_5V

Protected power domains

Sheet 2

Input clamp D7; eFuse U2 with UV/OV cut-off
OVP trip 5.28–5.47 V; LDO U4 (LT1763-3.3)

File: power.kicad_sch

MOTOR_RP/RN/LP/LN

Motor isolation and current limits

Sheet 3

Dual H-bridge U1 (DRV8411), enabled by MOTOR_EN_SAFE
Current limit 0.22–0.28 A per motor (R35, R36)
Schottky D5 isolates VM_MOTOR
U11 flags SYS_5V below 4.65 V on MOTOR_FAULT
Left motor J2, right motor J3

File: motor.kicad_sch

MOTOR_EN, HOST_HEARTBEAT, SERVO_REQUEST, PI_PROG_SWDIO/SWCLK/NRST

Independent safety supervisor

Sheet 4

Safety MCU U12 (STM32C011) on SAFE_3V3 (LDO U16)
Window watchdog U13; AND gates U14 (motor), U27 (servo)
Servo switch U15: SERVO_5V, 444–568 mA, 500 ms lease
Host line monitor U29; programming switch SW5

File: motor_safety.kicad_sch

MOTOR_EN_SAFE

MOTOR_FAULT

HOST_FAULT, HOST_ARMED

SENSOR_POWER_EN, SERVO_5V

MOTOR_FAULT

I2C_SDA/SCL, I2C_INT, US_TRIG/ECHO, SERVO_SIG

Protected external signals

Sheet 7

I2C repeaters U6 (expansion), U7 (LED driver)
I2C overvoltage protection U10
Sensor switch U30: SENSOR_5V, 108–154 mA
Trigger/servo buffer U8; echo/interrupt buffer U9

File: interfaces.kicad_sch

U3_SDA/SCL

Underlighting

Sheet 5

LED driver U3 (IS31FL3218), 6.3 mA per channel
Six RGB underlights RGB1–RGB6 on the bottom side

File: lighting.kicad_sch

EXP_I2C_SDA/SCL, EXP_INT_INPUT

Fixed expansion connectors

Sheet 6

I2C sockets X1, X3 (4-way, 1.0 mm pitch)
Header positions J4, JP1–JP5 (DNP)
Accessory rail EXP_3V3

File: expansion.kicad_sch

clamps

Connector-local ESD

Sheet 9

Clamp arrays U17–U26 (TPD4E05U06) and button clamps D8–D11 (TPD1E05U06) at each connector
Lines: EXP_I2C_SDA/SCL, EXP_INT_INPUT, TRIG_EXT, ECHO_INPUT, SERVO_EXT, SW_A_EXT–SW_Y_EXT and rails EXP_3V3, SENSOR_5V, SERVO_5V

File: esd.kicad_sch

clamps

TRIG_EXT, ECHO_INPUT, SERVO_EXT, SENSOR_5V

Buttons and indicators

Sheet 8

Buttons SW1–SW4 and indicators D1–D4
Ultrasonic sensor socket J1 on SENSOR_5V
Servo header J5 on SERVO_5V (DNP)

File: user_io.kicad_sch

LED_A/B/X/Y, SW_A/B/X/Y

Power rails

Rail	Source (sheet)	Sheets	Pad
PI_5V	host pins 2, 4	2	TP1
PI_3V3	host pins 1, 17	4, 7	TP4
SYS_5V	eFuse U2 (2)	2–5, 7	TP2
EXP_3V3	LDO U4 (2)	6, 7, 9	TP5
VM_MOTOR	diode D5 (3)	3	TP3
SAFE_3V3	LDO U16 (4)	3, 4, 7	TP39
SERVO_5V	switch U15 (4)	4, 7–9	TP45
SENSOR_5V	switch U30 (7)	7–9	–

Connectors

Ref	Function	Sheet
PI1	Host GPIO socket, 2x20	1
J2, J3	Left, right motor, 1x02	3
X1, X3	I2C sockets, 1x04	6
J4	I2C header, 1x05 (DNP)	6
JP1–JP5	I2C headers, 2x05 (DNP)	6
J1	Ultrasonic sensor, 1x04	8
J5	Servo header, 1x03 (DNP)	8

Schematic conventions

1. Sheets connect through global labels and power symbols; the sheet symbols carry no pins. Arrows on this page name the main nets between sheets.
2. A global label's shape gives the signal direction seen from its sheet; rectangular labels are passive.
3. Values name the fitted part; the MPN, Manufacturer and Selected_rating fields give the full specification.
4. Parts drawn crossed out carry the DNP attribute and are not fitted.

CERN-OHL-P-2.0

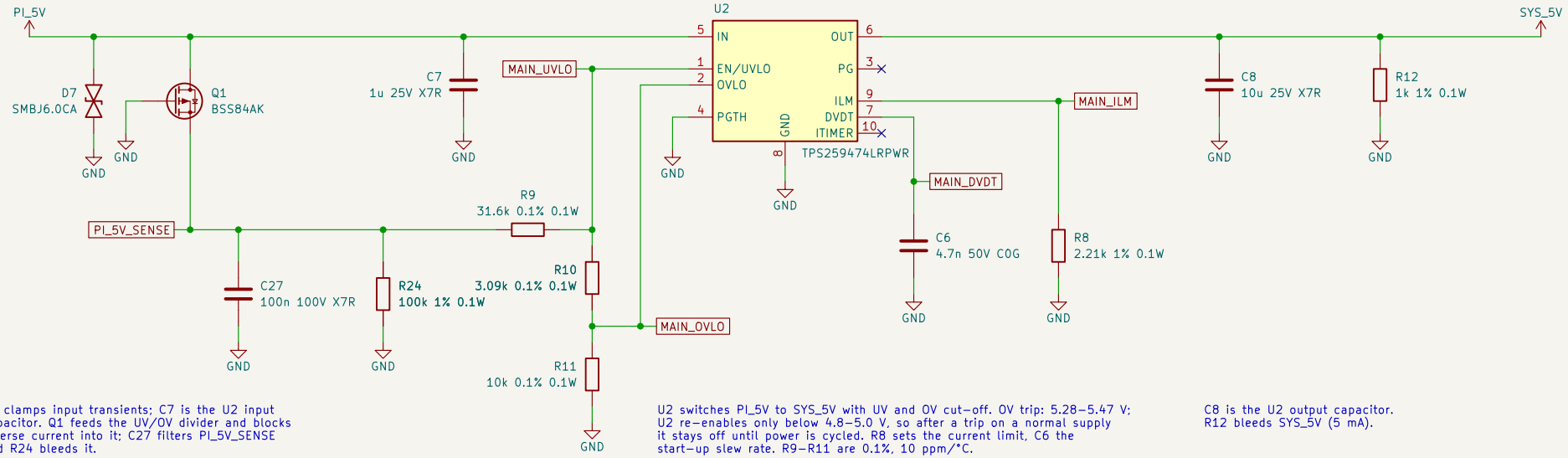
BisocM

Sheet: /
File: Keel.kicad_sch

Title: Keel R3 / Host interface

Size: A3	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 1/11

Input protection and eFuse (D7, U2)

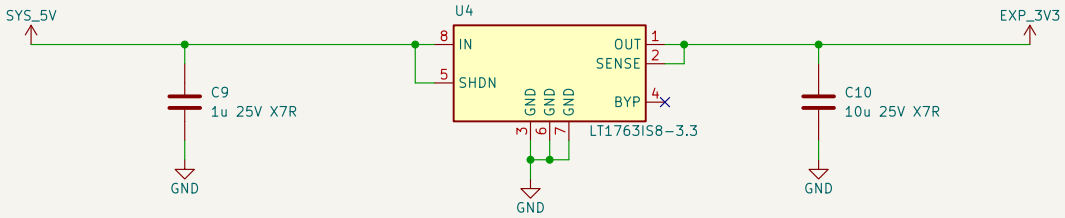


D7 clamps input transients; C7 is the U2 input capacitor. Q1 feeds the UV/OV divider and blocks reverse current into it; C27 filters PI_5V_SENSE and R24 bleeds it.

U2 switches PI_5V to SYS_5V with UV and OV cut-off. OV trip: 5.28–5.47 V; U2 re-enables only below 4.8–5.0 V, so after a trip on a normal supply it stays off until power is cycled. R8 sets the current limit, C6 the start-up slew rate. R9–R11 are 0.1%, 10 ppm/°C.

C8 is the U2 output capacitor. R12 bleeds SYS_5V (5 mA).

EXP_3V3 regulator (U4)



U4 makes EXP_3V3 for the expansion connectors on sheet 6 (Fixed expansion connectors), so accessories do not load the host 3.3 V rail. SHDN is tied to IN, so U4 runs whenever SYS_5V is up; SENSE is tied to OUT. C9 is the U4 input capacitor, C10 the output capacitor.

Notes

1. PI_5V: host 5 V from header pins 2 and 4 on sheet 1 (Host interface).
2. SYS_5V: eFuse-protected 5 V for U4, the motor supply and supervisor on sheet 3 (Motor isolation and current limits), the safety LDO and servo switch on sheet 4 (Independent safety supervisor), the LED driver and underlights on sheet 5 (Underlighting), and the sensor switch, I2C repeater and over-voltage protector on sheet 7 (Protected external signals).
3. EXP_3V3: 3.3 V for the expansion connectors on sheet 6 (Fixed expansion connectors).

CERN-OHL-P-2.0

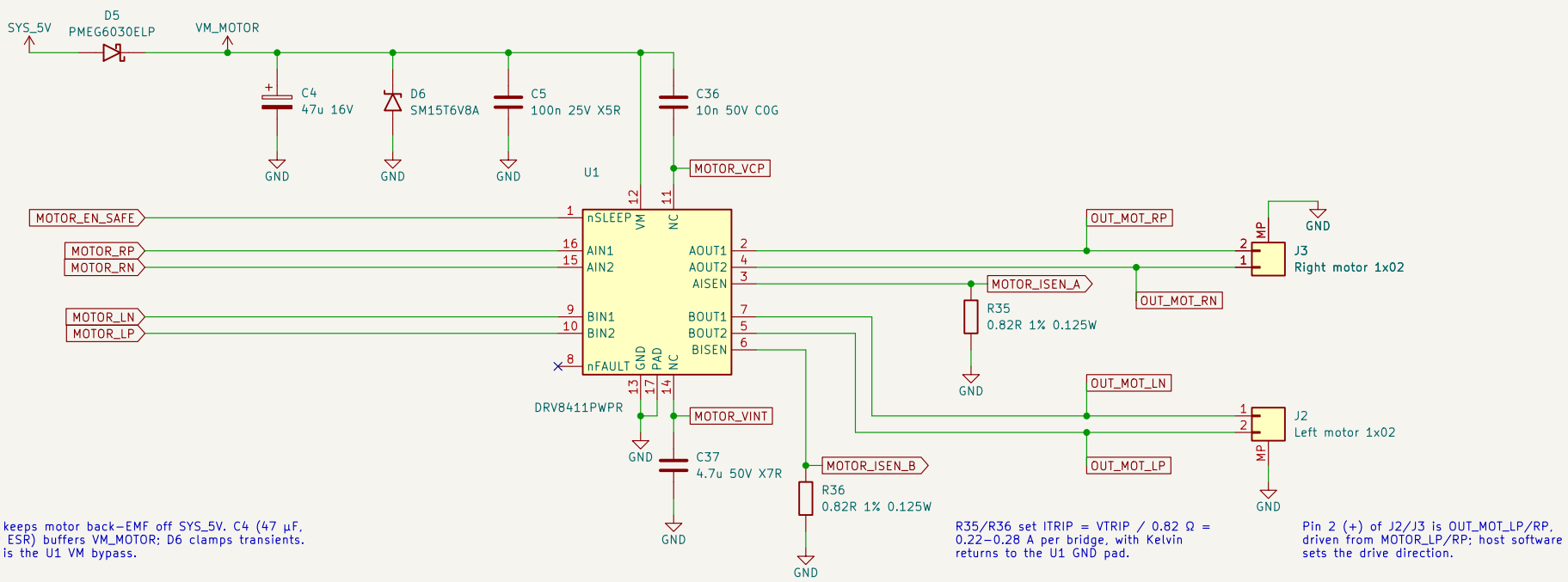
BisocM

Sheet: /Protected power domains/
File: power.kicad_sch

Title: Keel R3 / Protected power domains

Size: A4	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 2/11

Motor supply and dual H-bridge (D5, U1)

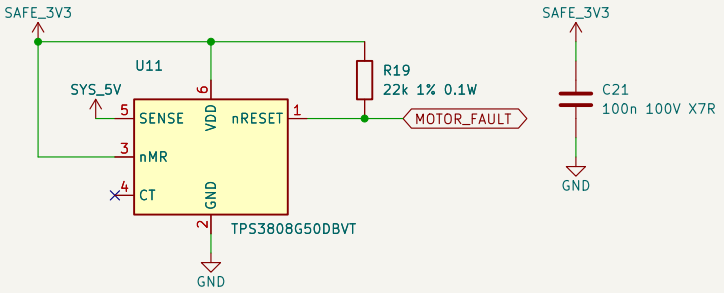


D5 keeps motor back-EMF off SYS_5V. C4 (47 µF, low ESR) buffers VM_MOTOR; D6 clamps transients. C5 is the U1 VM bypass.

R35/R36 set ITRIP = VTRIP / 0.82 Ω = 0.22–0.28 A per bridge, with Kelvin returns to the U1 GND pad.

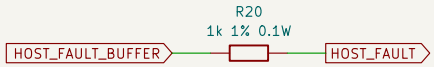
Pin 2 (+) of J2/J3 is OUT_MOT_LP/RP, driven from MOTOR_LP/RP; host software sets the drive direction.

SYS_5V supervisor (U11) and SAFE_3V3 bypass (C21)



U11 pulls MOTOR_FAULT low while SYS_5V is below 4.65 V and releases it 12–28 ms after recovery (CT open). R19 is the fault-bus pull-up. C21 is the U12 VDD bypass and sits at U12. U12 and C33, the U11 VDD bypass, are on sheet 4 (Independent safety supervisor).

READY status to host (R20)



R20 passes READY from status buffer U28 to host GPIO20, header pin 38. U28 runs from the host 3.3 V rail; RN6 pulls HOST_FAULT low. U28 and RN6 are on sheet 4 (Independent safety supervisor). HIGH = healthy (READY); LOW = fault, or the safety domain is unpowered.

Notes

1. U1 sleeps (nSLEEP low) unless gate U14 on sheet 4 (Independent safety supervisor) asserts MOTOR_EN_SAFE; sleep is the power-up default.
2. U1 nFAULT (pin 8) is left open: on the fault bus it would hold its own nSLEEP low through U14. U1 still protects itself (overcurrent retry, thermal shutdown, UVLO).
3. C36/C37 on U1 pins 11/14 (NC on DRV8411) provide VCP/VINT for the pin-compatible DRV8833PWPR.

CERN-OHL-P-2.0

BisocM

Sheet: /Motor isolation and current limits/
File: motor.kicad_sch

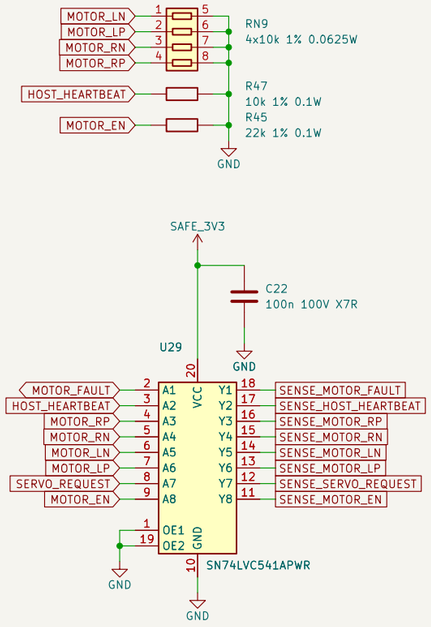
Title: Keel R3 / Motor isolation and current limits

Size: A4	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 3/11

Notes

1. Host request: MOTOR_EN (arming sequence), HOST_HEARTBEAT, SERVO_REQUEST and the PWM lines MOTOR_RP/RN/LP/LN reach U12 only through buffer U29.
2. MCU permit: U12 raises MCU_MOTOR_PERMIT or MCU_SERVO_PERMIT only after a valid arming sequence with a live heartbeat, and services watchdog U13 only while its own checks pass.
3. Hardware gates: U14 drives MOTOR_EN_SAFE (motor driver nSLEEP), U27 drives SERVO_POWER_EN (switch U15). Each is HIGH only while MOTOR_EN, its permit and MOTOR_FAULT are HIGH.
4. Fault bus: MOTOR_FAULT is open drain, LOW = fault, driven by U11, U30, U13 and U15. LOW turns both gates off and resets U12; actuators stay off until the host arms again.
5. Programming: SW5 in PROG disables U13 and clamps both permits LOW while U12 is halted.
6. No actuator permission survives a reset, a watchdog timeout or a recognized fault.
7. Safety MCU firmware: firmware/safety-mcu.
8. Motor driver U1 and supervisor U11: sheet 3 (Motor isolation and current limits).
9. Sensor switch U30: sheet 7 (Protected external signals).

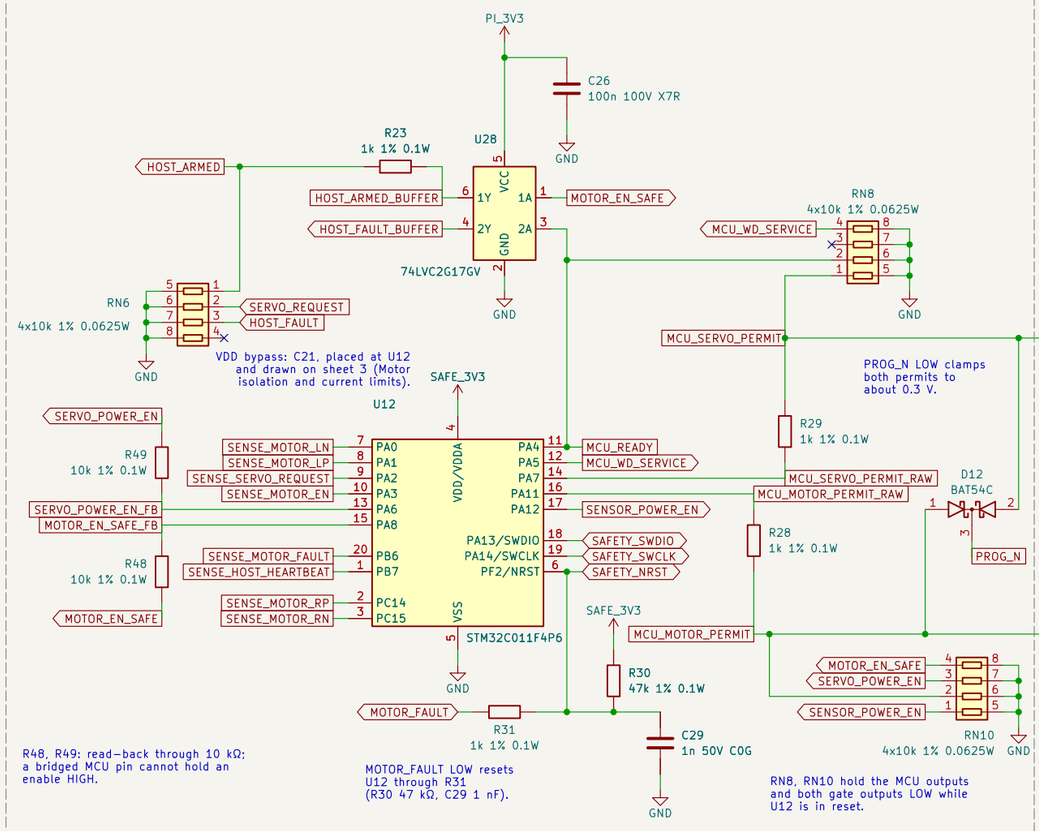
Host request buffer (U29)



U29: 5.5 V-tolerant inputs with Ioff, OE1/OE2 tied LOW.
RN9, R45, R47 and RN6 (pin 2) hold the host lines LOW while the host is absent or booting.

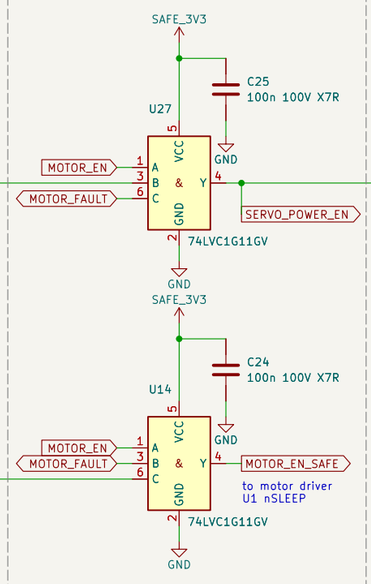
Safety MCU (U12) and host status (U28)

U28 runs from PI_3V3, so the host pins cannot back-power the safety domain.
HOST_FAULT_BUFFER HIGH = healthy (READY); it reaches HOST_FAULT through R20 on sheet 3 (Motor isolation and current limits).
HOST_ARMED HIGH = motor enabled (MOTOR_EN_SAFE).

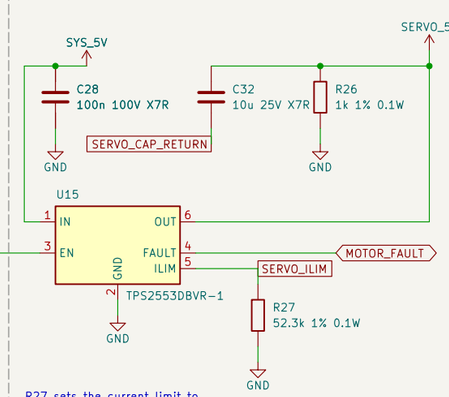


Enable gates (U27, U14)

Y = MOTOR_EN AND permit AND MOTOR_FAULT (HIGH = no fault).
No firmware in the off path: the host or the fault bus can always remove power.

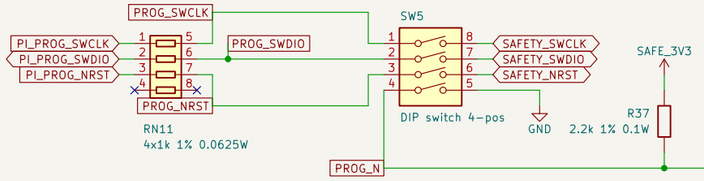


Servo power switch (U15)



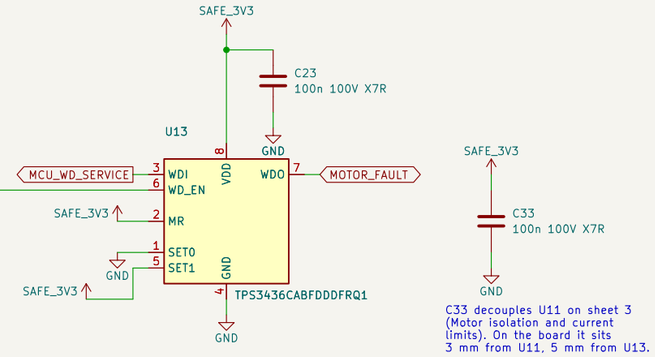
R27 sets the current limit to 444–568 mA: latch-off, FAULT pulls MOTOR_FAULT LOW.
C32 returns through R33 (2.2 Ω) on sheet 7 (Protected external signals) to damp hot-plug ringing.
R26 bleeds SERVO_5V.

Programming switch (SW5)



PROG (all poles ON): SWD and NRST connect to the host, U13 is disabled and D12 clamps both permits.
RUN (all poles OFF): the host is disconnected.
RN11: 1 kΩ in series with each host line.

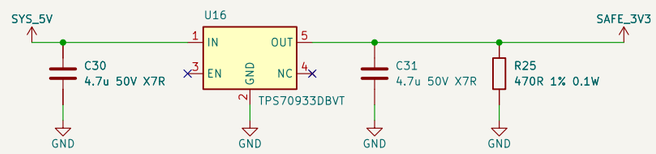
Window watchdog (U13)



SET1:SET0 = 1:0: closed window 0.8–1.2 ms, frame 102.4–153.6 ms, WDO LOW 22.5–27.5 ms. U12 services WDI every 50 ms. WD_EN = PROG_N (R37 pull-up): disabled only in PROG.

C33 decouples U11 on sheet 3 (Motor isolation and current limits). On the board it sits 3 mm from U11, 5 mm from U13.

SAFE_3V3 supply (U16)



U16 EN floats (internal pull-up): always on.
R25 holds SAFE_3V3 near 0 V when U16 is off.

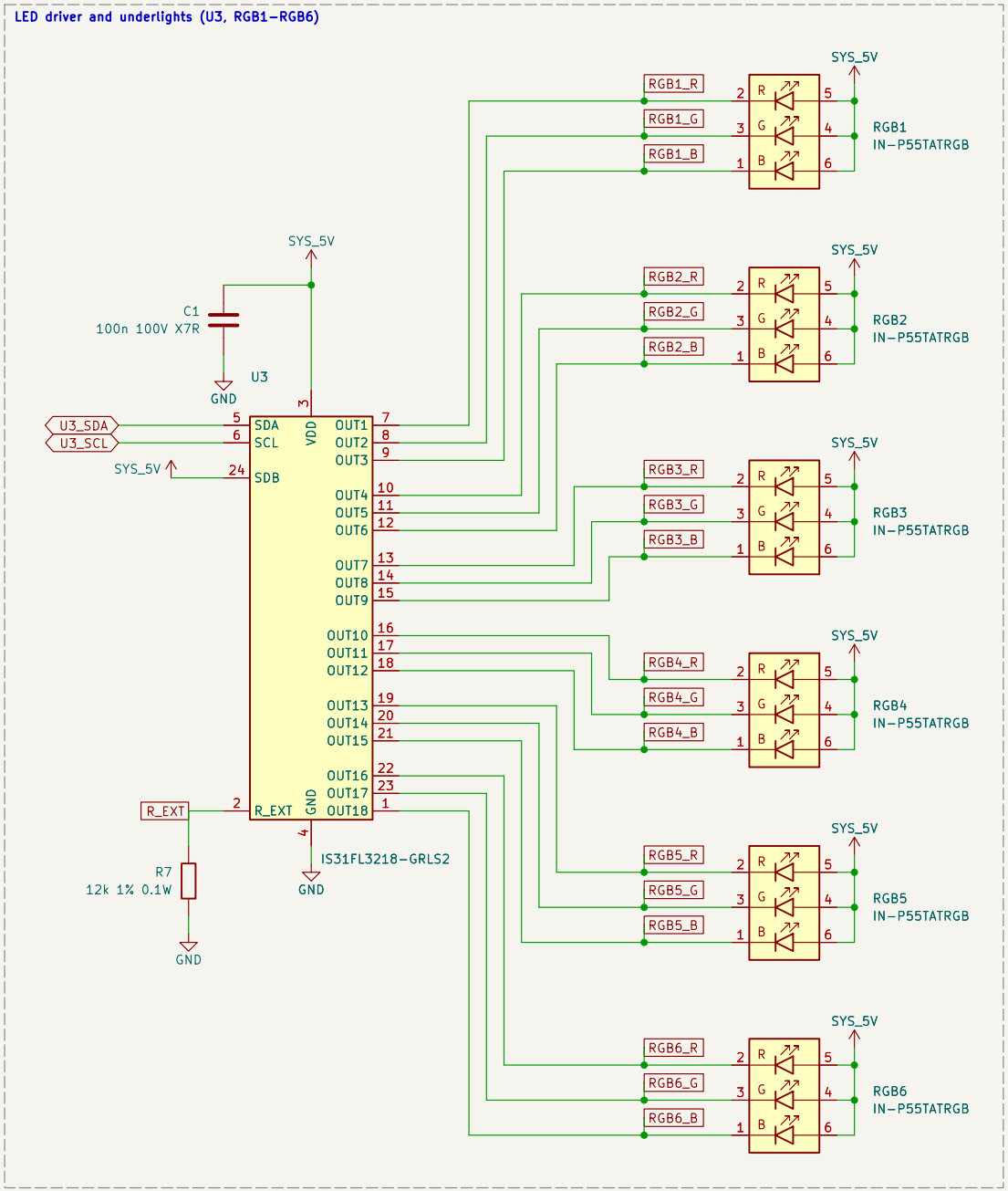
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BisocM

Sheet: /Independent safety supervisor/
File: motor_safety.kicad_sch

Title: Keel R3 / Independent safety supervisor

Size: A3	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 4/11



Notes

1. U3 has the fixed 7-bit I2C address 0x54. U3_SDA and U3_SCL come from repeater U7 on sheet 7 (Protected external signals); R38/R39 (1.8 kΩ) pull them up to SYS_5V.
2. SDB is tied to SYS_5V: U3 is switched on and off through register 0x00.
3. R7 (12 kΩ) sets the output current to 6.3 mA peak per channel; each channel has an 8-bit PWM register for its brightness.
4. RGB1–RGB6 are mounted on the bottom side, with their anodes on SYS_5V.

U3 registers

Register	*Function*
0x00	Shutdown: 0 = off, 1 = normal operation
0x01–0x12	PWM duty of OUT1–OUT18, 8 bits each
0x13–0x15	Output enables, six outputs per register
0x16	Update: write 0 to load PWM and enable data
0x17	Reset: write 0 to restore the defaults

Channel map

LED	*Outputs (R, G, B)*	*PWM registers*
RGB1	OUT1–OUT3	0x01–0x03
RGB2	OUT4–OUT6	0x04–0x06
RGB3	OUT7–OUT9	0x07–0x09
RGB4	OUT10–OUT12	0x0A–0x0C
RGB5	OUT13–OUT15	0x0D–0x0F
RGB6	OUT16–OUT18	0x10–0x12

CERN-OHL-P-2.0

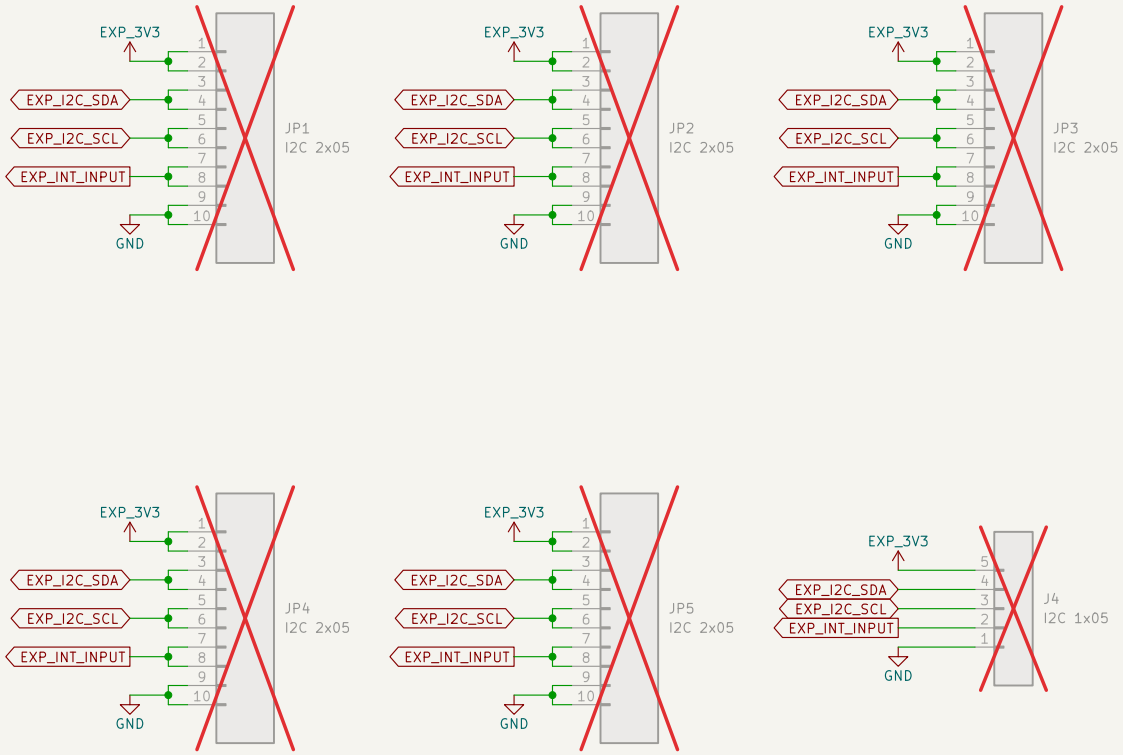
BisocM

Sheet: /Underlighting/
File: lighting.kicad_sch

Title: Keel R3 / Underlighting

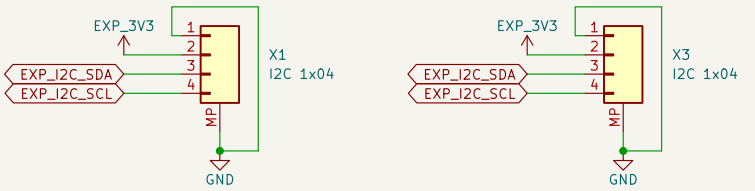
Size: A4	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 5/11

Expansion header positions (JP1–JP5, J4, DNP)



Each 2x05 position is two 1x05 rows 5.08 mm apart; pins 2n−1 and 2n are joined and carry one signal.

I2C sockets (X1, X3)



4-way 1.0 mm-pitch sockets; both mounting pins on GND.

Connector pinout

Net	JP1–JP5 (2x05)	J4 (1x05)	X1, X3 (1x04)
EXP_3V3	1, 2	5	2
EXP_I2C_SDA	3, 4	4	3
EXP_I2C_SCL	5, 6	3	4
EXP_INT_INPUT	7, 8	2	–
GND	9, 10	1	1, MP

Notes

1. EXP_I2C_SDA/SCL reach the host I2C bus through over-voltage protector U10 and repeater U6 on sheet 7 (Protected external signals).
2. EXP_INT_INPUT is a shared interrupt line: R17 (10 kΩ) pulls it up to EXP_3V3 and U9 buffers it for the host, both on sheet 7 (Protected external signals).
3. EXP_3V3 comes from LDO U4 on sheet 2 (Protected power domains).
4. Each connector has a TPD4E05U06 clamp beside it on sheet 9 (Connector-local ESD): U17–U21 at JP1–JP5, U25 at J4, U22 at X1, U23 at X3.

CERN-OHL-P-2.0

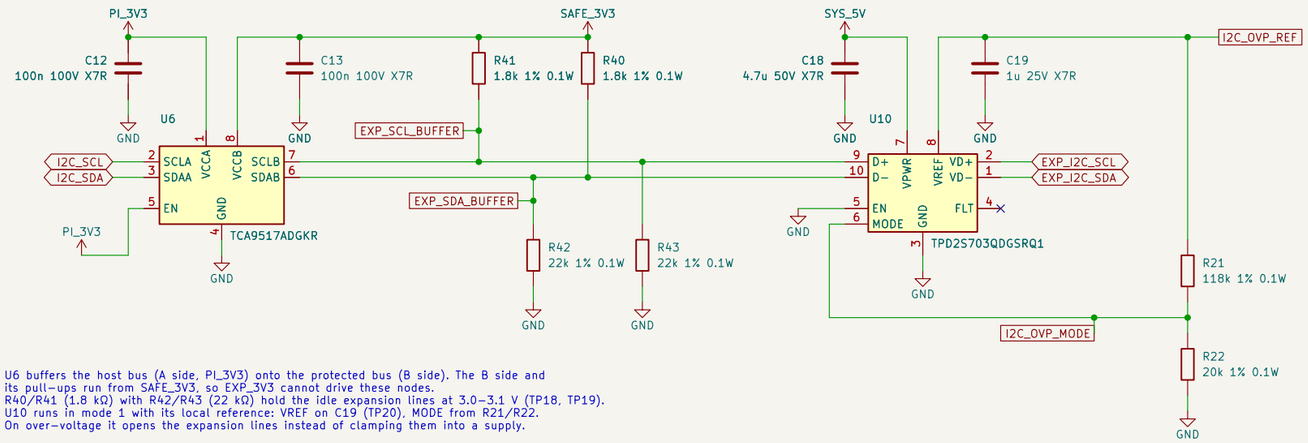
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Sheet: /Fixed expansion connectors/
File: expansion.kicad_sch

Title: Keel R3 / Fixed expansion connectors

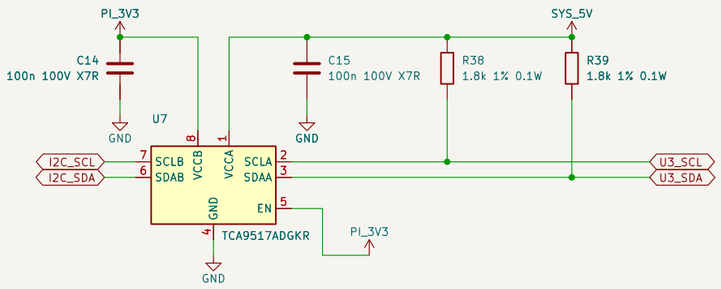
Size: A4	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 6/11

Expansion I2C buffer and over-voltage switch (U6, U10)



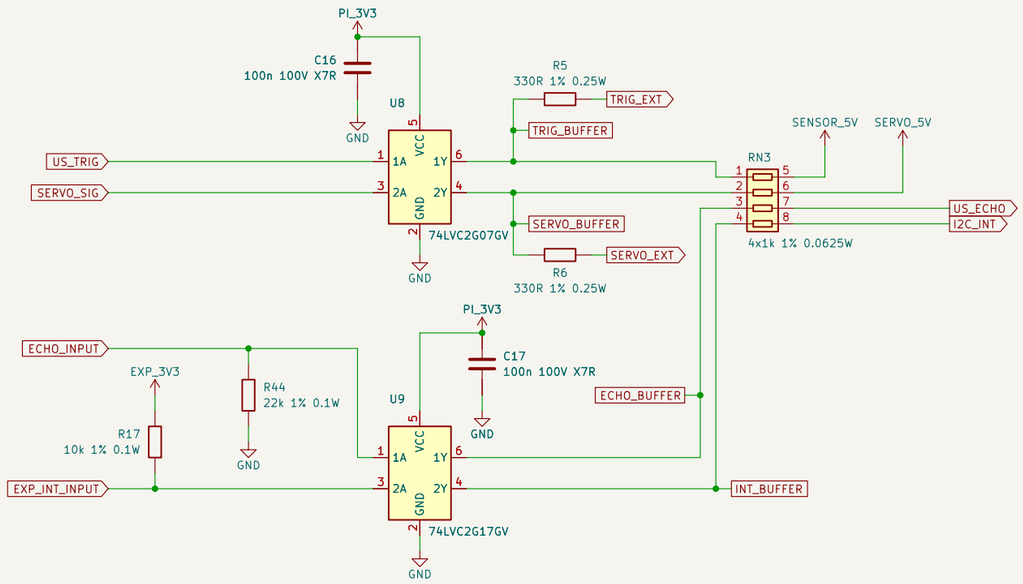
U6 buffers the host bus (A side, PI_3V3) onto the protected bus (B side). The B side and its pull-ups run from SAFE_3V3, so EXP_3V3 cannot drive these nodes.
R40/R41 (1.8 kΩ) with R42/R43 (22 kΩ) hold the idle expansion lines at 3.0–3.1 V (TP18, TP19).
U10 runs in mode 1 with its local reference: VREF on C19 (TP20), MODE from R21/R22.
On over-voltage it opens the expansion lines instead of clamping them into a supply.

LED-driver I2C buffer (U7)



U7 is connected the reverse way to U6: host bus on the B side (PI_3V3), 5 V LED-driver bus on the A side (SYS_5V, R38/R39 1.8 kΩ pull-ups).
The LED driver U3 is on sheet 5 (Underlighting).

Ultrasonic, servo and interrupt buffers (U8, U9, RN3)



U8 open-drain outputs are pulled up through RN3 (1 kΩ) to the switched rails SENSOR_5V and SERVO_5V, so a disabled accessory is not back-powered through its signal line.
R5/R6 (330 Ω) protect the outputs against a 5.5 V fault on the connector.
U9 Schmitt inputs tolerate 5.5 V; RN3 elements 3 and 4 are 1 kΩ series resistors to the host.
R44 holds ECHO_INPUT low and R17 holds EXP_INT_INPUT high when nothing drives them.

Sensor 5 V switch and capacitor returns (U30, R33, R34)

U30 switches SENSOR_5V from SYS_5V. EN is active high, from the safety MCU U12.
U30 latches off after an overload; R32 (210 kΩ) sets the limit to 108–154 mA.
FAULT drives the shared open-drain MOTOR_FAULT bus. R46 discharges SENSOR_5V when off.
R34 and R33 (2.2 Ω, pulse-rated) damp hot-plug ringing in the returns of C35 and of the servo-rail capacitor C32 on sheet 4 (Independent safety supervisor); they carry no DC load current.

Notes

1. The ESD clamps of the connector-side lines sit at the connectors, on sheet 9 (Connector-local ESD).
2. SAFE_3V3 (safety-domain rail), SERVO_5V (switch U15), SENSOR_POWER_EN (MCU U12) and SERVO_CAP_RETURN (C32) come from sheet 4 (Independent safety supervisor).
3. Expected readings at TP18–TP20 are on sheet 10 (Signal and programming probe access).

CERN-OHL-P-2.0

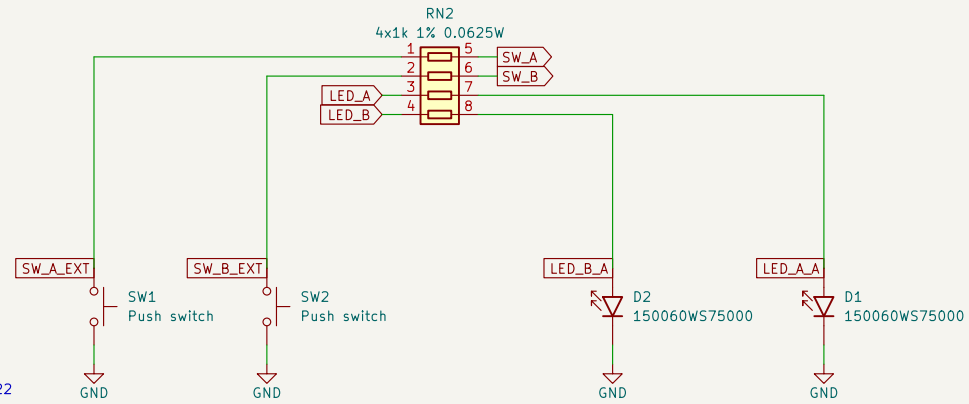
BisocM

Sheet: /Protected external signals/
File: interfaces.kicad_sch

Title: Keel R3 / Protected external signals

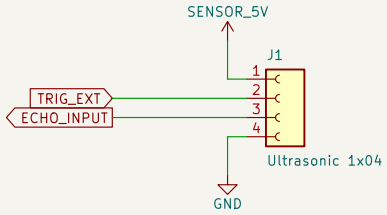
Size: A3	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 7/11

Buttons A, B and indicators (SW1, SW2, D1, D2)



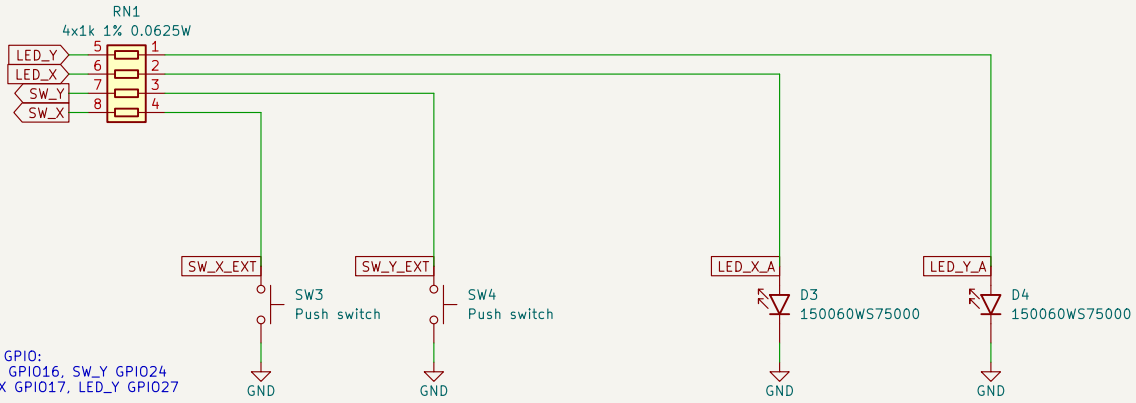
Host GPIO:
SW_A GPIO5, SW_B GPIO6
LED_A GPIO23, LED_B GPIO22

Ultrasonic sensor socket (J1)



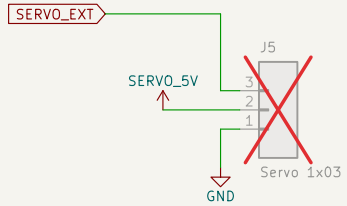
SENSOR_5V: switch U30 (108–154 mA limit).
TRIG_EXT from buffer U8 via R5; ECHO_INPUT to buffer U9.
U30, U8 and U9 are on sheet 7 (Protected external signals).
Clamp U24 on pins 1–3: sheet 9 (Connector–local ESD).

Buttons X, Y and indicators (SW3, SW4, D3, D4)



Host GPIO:
SW_X GPIO16, SW_Y GPIO24
LED_X GPIO17, LED_Y GPIO27

Servo header (J5, DNP)



SERVO_5V: switch U15 (444–568 mA limit) on sheet 4 (Independent safety supervisor).
SERVO_EXT from buffer U8 via R6 on sheet 7 (Protected external signals).
Clamp U26 on pins 2 and 3: sheet 9 (Connector–local ESD).

Notes

1. The buttons are active low with no external pull–ups: the host enables its internal GPIO pull–ups.
2. Each button and indicator line runs through a 1 kΩ element of RN1 or RN2. It limits the current into a host pin driven against a pressed button, and sets the indicator current below 1 mA from 3.3 V.
3. D8–D11 on sheet 9 (Connector–local ESD) clamp SW_A_EXT, SW_B_EXT, SW_X_EXT and SW_Y_EXT.

CERN–OHL–P–2.0

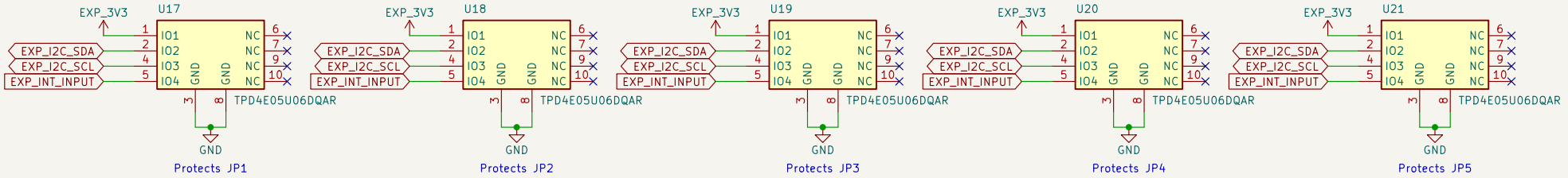
BisocM

Sheet: /Buttons and indicators/
File: user_io.kicad_sch

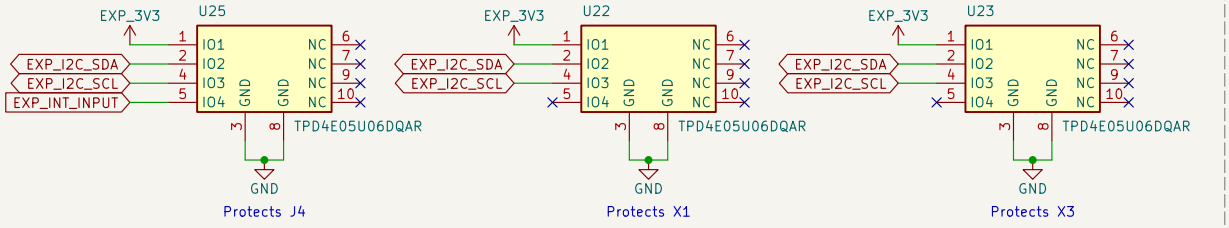
Title: Keel R3 / Buttons and indicators

Size: A4	Date: 2026–11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 8/11

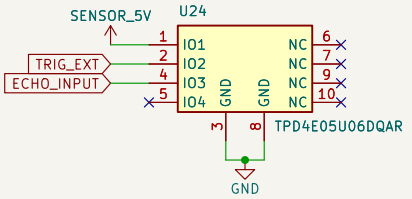
Expansion header clamps (U17–U21)



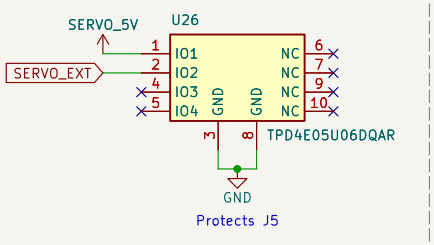
J4 and I2C socket clamps (U25, U22, U23)



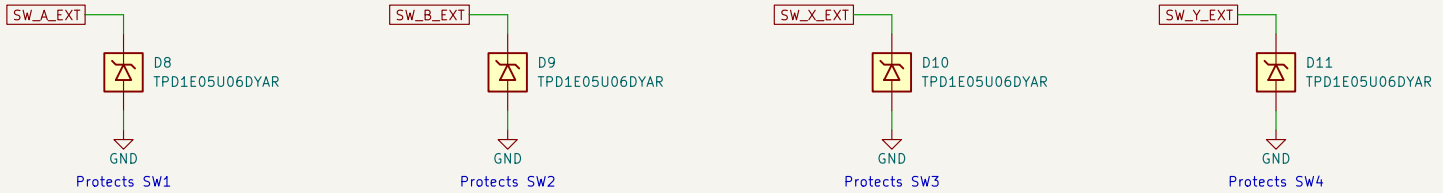
Ultrasonic sensor socket clamp (U24)



Servo header clamp (U26)



Push-button clamps (D8–D11)



Notes

1. Each clamp sits beside the connector or switch it protects, with a short, direct return to GND.
2. IO1 of each array clamps the connector supply pin (EXP_3V3, SENSOR_5V or SERVO_5V).
3. The clamps absorb ESD transients only. Sustained faults are handled downstream by the series resistors, current-limited supplies and the over-voltage protector on sheet 2 (Protected power domains), sheet 4 (Independent safety supervisor), sheet 7 (Protected external signals) and sheet 8 (Buttons and indicators).
4. TPD4E05U06 pins 6, 7, 9 and 10 are NC flow-through pads; spare IO channels are left open.

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BisocM

Sheet: /Connector-local ESD/
File: esd.kicad_sch

Title: Keel R3 / Connector-local ESD

Size: A4	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 9/11

POWER row

Pos	Net	Ref	Silkscreen	Expected reading
1	SYS_5V	TP2	2 SYS5V	Within 60 mV of TP1 with U2 on; 0 V after a trip
2	GND	TP6	6 GND	Row ground return
3	PI_5V	TP1	1 PI5V	5.0–5.1 V; below 5.2 V (U2 OVP trip 5.28 V min)

MOTOR row

Pos	Net	Ref	Silkscreen	Expected reading
1	MOTOR_ISEN_B	TP13	13 ISB	Left motor shunt R36; chops at 180–230 mV
2	MOTOR_ISEN_A	TP12	12 ISA	Right motor shunt R35; chops at 180–230 mV
3	GND	TP25	25 GND	Row ground return, sheet 11 (Local probe ground returns)
4	VM_MOTOR	TP3	3 VM	TP2 minus the D5 drop (0.3–0.4 V while driving)

SERVO row

Pos	Net	Ref	Silkscreen	Expected reading
1	SERVO_5V	TP45	45 S5V	5 V only during a servo burst (500 ms max)
2	GND	TP47	47 GND	Row ground return, sheet 11 (Local probe ground returns)

SWD row

Pos	Net	Ref	Silkscreen	Expected reading
1	SAFE_3V3	TP39	39 3V3S	3.234–3.366 V; probe VTref only, never feed
2	GND	TP40	40 GND	Row ground return
3	SAFETY_NRST	TP41	41 RST	U12 NRST; held LOW while MOTOR_FAULT is LOW
4	SAFETY_SWCLK	TP43	43 CLK	SWD clock, U12 PA14–BOOT0
5	SAFETY_SWDIO	TP42	42 DIO	SWD data, U12 PA13

Set SW5 to PROG (all ON) before attaching an SWD probe.

STATUS row

Pos	Net	Ref	Silkscreen	Expected reading
1	MCU_WD_SERVICE	TP44	44 WDI	U12 to U13 WDI; falling edge every 50 ms
2	PI_3V3	TP4	4 3V3P	3.3 V with the host running; 0 V without it
3	GND	TP26	26 GND	Row ground return, sheet 11 (Local probe ground returns)
4	MOTOR_FAULT	TP10	10 FLT	Fault bus: HIGH = healthy, LOW = fault
5	HOST_ARMED	TP21	21 ARM	HIGH = motors armed (to host GPIO21)
6	MOTOR_EN	TP9	9 MEN	Host motor request (GPIO26); 22 kΩ pull-down R45
7	SERVO_POWER_EN	TP46	46 SVEN	U27 output; HIGH only during a servo burst

I2C row

Pos	Net	Ref	Silkscreen	Expected reading
1	I2C_SDA	TP7	7 SDA	Host I2C data (GPIO2), host pull-up
2	I2C_SCL	TP8	8 SCL	Host I2C clock (GPIO3), host pull-up
3	GND	TP24	24 GND	Row ground return, sheet 11 (Local probe ground returns)
4	EXP_I2C_SDA	TP18	18 ESDA	Expansion side of U10; idle 3.0–3.1 V
5	EXP_I2C_SCL	TP19	19 ESCL	Expansion side of U10; idle 3.0–3.1 V
6	I2C_OVP_REF	TP20	20 OREF	U10 OVP reference, 3.25–3.65 V
7	EXP_3V3	TP5	5 E3V3	3.3 V from U4 whenever SYS_5V is up

Notes

1. Bare 1.5 mm ENIG pads on the bottom side, without paste or a fitted part; they are not in the BOM.
2. All pads sit on the 2.54 mm grid of the host header PI1, so one pogo-pin plate reaches all 28 pads.
3. Pos counts along the row from its silkscreen label (west to east in top view).
4. Each row has its own GND pad; TP6 and TP40 here; TP24, TP25, TP26 and TP47 are drawn on sheet 11 (Local probe ground returns).
5. A bench supply goes into TP1 (+) and TP6 (GND), never into TP39, and never with a powered host fitted.
6. Programming and test fixtures keep the actuator permits inactive.

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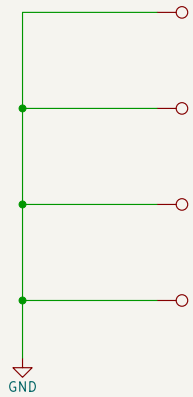
Sheet: /Signal and programming probe access/
File: testpoints.kicad_sch

Title: Keel R3 / Signal and programming probe access

Size: A3	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 10/11

Row GND pads

Row	Pos	Net	Ref	Silkscreen	Ground return for
MOTOR	3	GND	TP25	25 GND	TP13, TP12, TP3
SERVO	2	GND	TP47	47 GND	TP45
STATUS	3	GND	TP26	26 GND	TP44, TP4, TP10, TP21, TP9, TP46
I2C	3	GND	TP24	24 GND	TP7, TP8, TP18, TP19, TP20, TP5



Notes

1. Bare 1.5 mm ENIG pads on the bottom side, without paste or a fitted part; they are not in the BOM.
2. Each pad sits in its probe row on the 2.54 mm grid, beside the row signals, for a short return loop.
3. Pos counts along the row from its silkscreen label (west to east in top view), as on sheet 10 (Signal and programming probe access).
4. The POWER and SWD row GND pads (TP6, TP40) are drawn with their rows on sheet 10 (Signal and programming probe access).

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BisocM

Sheet: /Local probe ground returns/
File: ground_probes.kicad_sch

Title: Keel R3 / Local probe ground returns

Size: A4	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 11/11